

Requirements for the Data Capturing (Camera)

Donnerstag, 29. Mai 2025 15:57

1. Placement of the camera:

Mount the camera Infront the person so that it can replicate the dataset used for the camera pipeline. Example is shown below:



- 1. Each participant must sit for 5 videos (approx 10 mins each) with different sleepiness scores assigned at the end by the participant themself (Recommendation: Since we have 5 participants , each participant will record 5 videos I.e. 5x5=25 videos. And each participant must record all the videos in one session to observe realistic drowsiness behavior)
(0(Extremely alert) to 9 (Very sleepy, great effort to keep awake, fighting sleep))

2. Avoid the following false positives that can actually cause trouble in feature extraction



Avoid wide smiles

Avoid nail biting



Avoid Putting hand on face after yawn



Avoid making camera adjustment during data capturing

3. Following is the KSS questionnaire template:

<https://docs.google.com/forms/d/e/1FAIpQLScmusV2nkdHw4CKKwrWHAc7Hi69GEF1Qc8j3IMtV5tALDdXw/viewform?usp=header>

KSS	Description	Blink Count in a minute	Yawn Rate (per 5 min)	Lateral Offset	Steering Angle
1–3		10–15	0	Not entering the opposite lane and avoiding high correction of lateral position.	Correcting after every 30 sec.
4–6	Moderate Fatigue	15–18	1	1 or 2 times entering the Solid-Solid Line type.	Correcting after every 12-15 sec.
7–9	Drowsy	> 18	> 1–2	9 or 10 times entering the Solid-Solid Line type.	Correcting after every 5 sec.

Reference for the observer (Approximate from Literature)

KSS	Description	Blink Rate (blinks /min)	Yawn Rate (per 5 min)	PERC LOS (%)	Head Nodding (nods/min)	SDLP (cm)	Steering Entropy	Lane Keeping Ratio	Lane Departure (per 5 min)	SRR (reversals/min)	Key References
1–3	Alert	10–15	0	< 15%	0	< 20	0–1	> 0.85	1.5–2.5	25–35	[1][2][3][4][5]
4–6	Moderate Fatigue	15–18	1	15–30%	1–2	20–35	1–3	0.7–0.85	2.5–3.5	15–25	[2][3][5][6]
7–9	Drowsy	> 18	> 1–2	> 30%	> 2	> 35	> 3	< 0.7	> 3.5	< 15	[2][4][6][7][8]

[1] PERCLOS (Percentage of Eye Closure)

[Wierwille & Ellsworth \(1994\) – Evaluation of PERCLOS for drowsiness detection](#)

[2] Blink Rate and Yawn Frequency

[Picot et al. \(2009\) – Eye behavior and EEG for fatigue detection](#)

[3] Infrared Reflectance Oculography and KSS

[Johns et al. \(2000\) – Infrared reflectance oculography vs KSS](#)

[4] Standard Deviation of Lane Position (SDLP)

[Dinges et al. \(1998\) – SDLP and KSS relationship](#)

[5] Lane Departure and Lane Keeping

[Vicente et al. \(2015\) – Lane behavior during fatigue](#)

[6]/[8] Steering Entropy and Steering Reversal Rate (SRR)

[Dong et al. \(2011\) – Steering entropy and inattention](#)

[Liang et al. \(2007\) – Glance behavior and crash risk prediction](#)

[7] Steering Variability

[Brown et al. \(2006\) – SRR and fatigue modeling](#)

- Image size (frame) must be of 512x424 in order to replicate the dataset (Signal Processing Team)
- All frames must be extracted and sent to us (No skipping of frames required)
- No cropping of picture required (Signal Processing team)

Requirements for the Data Capturing (Lane Pos & Steering Angle)

- Use same car and same map to record everything
- Keep a constant Speed only give the control of direction to the user.
- Observe the output with different weather conditions (Recommendation: It should return same output in case it doesn't we need to consider that while feature extraction)
- Use Tesla Model 3 as a car in the Carla
- Steering Angle**, What is the max and min of steering angle ? What's its unit rad ?
- Lateral offset and lane ID** and **is vehicle in intersection or not** is required for lane position
- Which map are we going to use for Carla simulation (Recommendation: Use the curved map for better and with 2 lanes on each side)

8. It's hard to maintain the lane position in the Carla (Recommendation: data capturing volunteer must do the trail before the actual data capturing)
9. Lane and steering data must be up to 3 decimal precision(example:0.004)
10. What is the frequency of recording from the CARLA for both the steering angle and the lateral offset?
11. Data must be captured in first person view FPV, following is the reference:
 - a. ''''



This is the image from CARLA.